

Supplementary Materials to the manuscript of Efimenko et al. "NeMu: A Comprehensive Pipeline for Accurate Reconstruction of Neutral Mutation Spectra from Evolutionary Data"

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SUPPLEMENTARY NOTES AND FIGURES

Supplementary Note 1. Data filtering

Data sampling. The primary objective of our NeMu pipeline is to extract the neutral (or near neutral) component of the species-specific mutational spectrum. To achieve this, we emphasize the importance of the use of synonymous substitutions (and some missense mutations also due to the evidence that at least some missense mutations may be neutral also (1–3)) in protein-coding sequences with clearly defined orthological relationships. Alignments that combine orthologs and paralogs can result in inaccurate phylogenetic reconstruction when using a simple substitution model and, consequently, a distorted mutational spectrum. Furthermore, paralogs are particularly susceptible to the fixation of substitutions through positive selection, a phenomenon stemming from their functional divergence. This can introduce additional biases into the reconstructed spectra (4–9). This requirement holds particular significance for the manual approach, whereas the automatic method, applied to mtDNA of *metazoa*, inherently utilizes natural well-defined mitochondrial orthologous groups. It is important to note that automatic sample construction method from MIDORI2 (10) mtDNA database have a simple technical insight enabled us to be sure that in the entry we still have one-to-one strict orthologs: the list of retrieved mitochondrial sequences (which are unique/non-duplicated in mtDNA) is automatically analyzed from top, best homologous to query, to bottom, last homologous to query, and analysis stops when the sequence belongs to other species found (this sequence used as an outgroup).

An essential factor in our analysis is the completeness of the mutational spectra, which is influenced by the number of substitutions observed in the analyzed dataset. This is particularly challenging for species-specific data. To ensure basic phylogenetic tree resolution, we require a minimum of four unique (nonredundant) sequences in the species-specific ingroup. However, this threshold is not always adequate for constructing complete mutational spectra, which consist of 192 components (calculated as 12 possible directions of simple nucleotide substitutions multiplied by 16 combinations of two neighbouring nucleotides). This problem is particularly noticeable when these four sequences exhibit a low count of synonymous substitutions. Therefore, at the step of the sample construction and especially in the case of species-specific data sampling methodology, it's crucial to balance the number of included species against the comprehensiveness of the species-specific spectra, ensuring a thorough and accurate representation of the mutational landscape. Conducting an analysis with an expanded data set is recommended to ensure there is enough data to reconstruct deep mutational spectra. However, our approach to constructing mutation spectra is based on phylogenetics rather than population models, which is why we don't recommend using complete within-population data for the NeMu pipeline (for evaluating mutation spectra based on population data, other software is available (11)).

The manual method of sample construction (e.g., set of aligned protein coding sequences as input) allows a freedom for user to build a mutational spectrum for other than species taxonomic levels, ranging from specific populations to broader categories like genus, families or orders. Moreover in the manual method any protein coding gene of nuclear, bacterial, mitochondrial and plastid genomes can be used. If possible, it is recommended that the sequence set be filled in by strict orthologs in this sample construction method. On the same time, we cannot recommend this method of sample construction to a datasets having synonymous substitution saturation (e.g., the rank of taxonomic class or higher) because of the much greater ambiguity in the classification of neutral (or nearly neutral) alignment positions with synonymous fixations at the deep inner tree branches (12, 13). To construct a mutational spectrum for higher taxonomic rank we recommend procedures described in our analysis of mtDNA mutation spectra of *Chordata* species (14). The manual method of sample construction is often exacerbated by unnoticed and not trivial difficulties: it is important for the user to make sure that the pairwise distance distribution between the aligned sequences is uniform (this means that it should not be multimodal), otherwise, the mutations on the longest tree branches, which correspond to outlayer branch distances, could cause a significant bias in the mutational spectrum (15). The general solution of this problem we provided in the next, phylogenetic, stage of data processing. However, we recommend the users to pre-check the dataset they upload to avoid automatic removal of sequences from the sample.

Following the construction of the sample, in the pipeline we align the protein coding sequences of the sample sequence set by MAFFT v7.505 (16, 17), which is utilized to quickly align translated protein sequences and then translate this alignment to codons using reportGapsAA2NT MACSE module. Also, we allow advanced users to enable MACSE v2.06 aligner (18) instead of MAFFT to run codon alignments directly.

Phylogeny and site-to-site alignment variability. For phylogenetic reconstruction we used IQ-TREE 2.2.0 (19). We chose this software because of the fast computation speed and flexibility in the evolutionary model setting. This software enables the straightforward application and evaluation of both time-reversible and non-reversible evolutionary Markov models in data analysis (20, 21). In situations characterized by extremely limited evolutionary timeframes, particularly in species-specific analysis of non-model organisms with low number of closely related sequences, non-reversible Markov models are the only models that accurately reflect sequence evolution (22, 23). Therefore, non-reversible Markov models are the default choice in our pipeline. The automatic re-rooting of phylogeny and other classic manipulations of phylogeny topology is conducted by Newick tools 1.6.0 package (24). The pre-final step in quality control prior to mutational spectra reconstruction is to detect branching anomalies. We require the outgroup branch length be longer than zero. If the length of the outgroup branch is zero, the pipeline raises an error since the outgroup may belong to ingroup (due to potential problems in sequence annotation in nucleotide databases) or there's a problem in the taxonomic definition of a given species. Under the manual mode of sequence preparation it is highly important to be sure that there are no obvious sub-trees or long branches in the reconstructed phylogeny. To verify this, we use the TreeShrink software (25), which implements an algorithm to find tree branches that could be removed to minimize the sum of branch lengths. TreeShrink is recommended by default because of the high frequency of various database errors (e.g., sequencing and/or annotation errors).

The final quality control step is masking of conservative sites, including synonymous ones, for subsequent reconstruction of the mutational spectra. The main assumption behind this is that conservative synonymous sites can be the result of a strong purifying selection and thus should be eliminated from the alignment. If the user wants to exclude negative selection for a given subset of alignment dataset (not recommended) it is possible to deactivate this option or run both of them in order to compare results. To determine whether the alignment sites are possible subject to negative natural selection in the simplest case, it is enough to analyze the variation of the alignment sites. To do this, we use the variability class prediction of each alignment site based on the standard evaluation of rate heterogeneity (gamma distribution) across sites, implemented in IQ-TREE 2.2.0 (26, 27). For example, on Figure S1A we show a rate heterogeneity characteristic for Hominida mtDNA genes. We obtained this estimate using both human data (GenBank database) and apes data (Pan, Gorilla and Pongo, species listed in Supplementary Table 4; GenBank database and (28)) because of the likely ineffectiveness of substituting all possible neutral mutations over the course of human evolution (coalescence time of human mtDNAs including neanderthal and Denisovan mtDNAs is less than 1 MYA). Figure S1B demonstrates a rate heterogeneity characteristic for Mus genus (*Mus musculus* data from GenBank database supplemented by other Mus species listed in Supplementary Table 4)

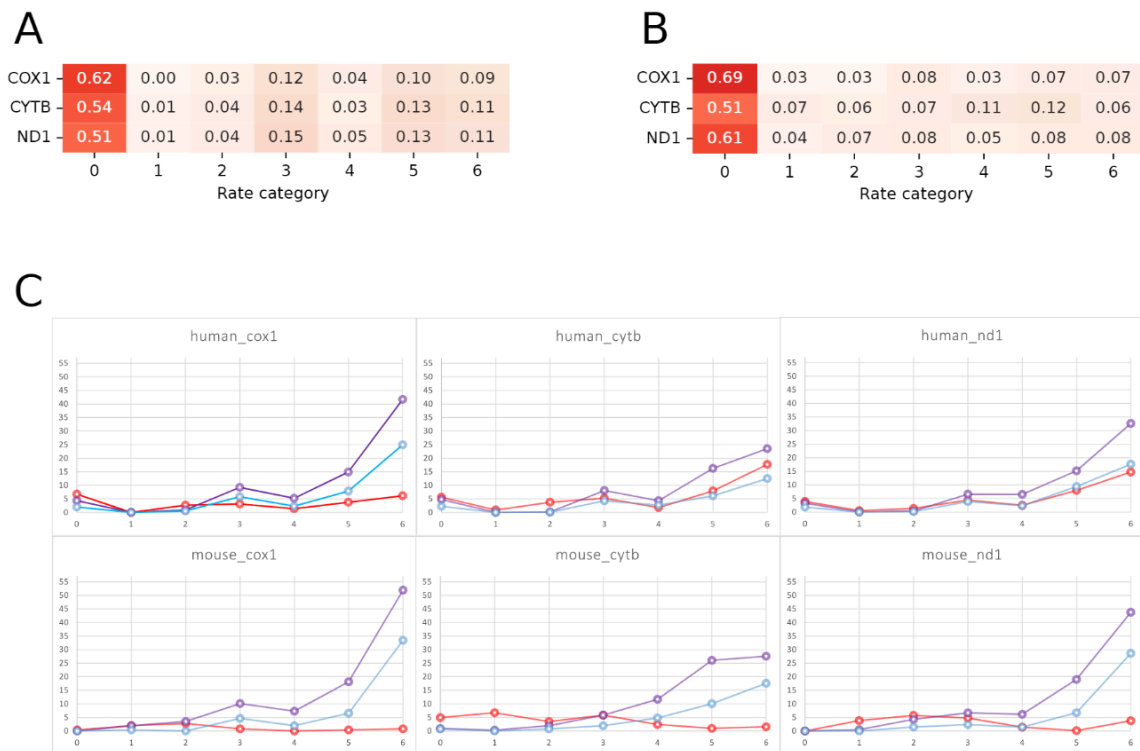


Figure S1. Variability of alignment sites in human and mouse mitochondrial genes. The fraction of alignment sites characterizing by 0 - 6 rate heterogeneity categories for 3 human (A) and 3 mouse (B) mitochondrial genes. Seven rate categories (from 0, totally conservative, to 6, totally variable) of gamma distribution were used in computations. (C) The percent (Y-axes) of substitution types and rate heterogeneity classes (X-axes) across alignment sites of human and mouse mitochondrial genes. Three substitution types were shown by different colors: total synonymous substitutions by purple, four-fold synonymous substitutions by blue, and non-synonymous substitutions by red.

Supplementary Note 2. Fast and accurate reconstruction of the ancestral states probabilities

We have focused our main efforts on reconstructing the evolutionary history of DNA as accurately as possible. In doing so, first of all, we have a possibility for the user to exclude from consideration all mutations localized on the terminal branches of phylogenetic trees, because these mutations may originate from sequencing artifacts or some kind of harmful somatic events (the choice to exclude terminal branches from the analysis depends on the accuracy of the data is monitored by the user). Secondly, we have based our ancestral state reconstruction logic on the simplified “uncertainty principle” proposed earlier, which essence is the inability to evaluate simultaneously and precisely the rate of evolution of DNA sequences and ancestral states on a long and short branches of phylogenetic tree (the higher the rate of evolution on any branch of the phylogenetic tree is, the less likely it is to determine ancestry, and hence the types of nucleotide substitutions, at the extremities of that branch; and vice versa, the more the ancestral status is determined at the tree nodes, the less reliable the rate of evolution can be determined) (29). It is the default approach to restore ancestral states on the comparative species scale; also it is possible for the user to use it in any other situation. In order to balance evolutionary paths in terms of rates of evolution and reconstructability of ancestral states (biologically meaningful or having sufficient information about the types and number of nucleotide substitutions; see illustrative Appendix S1 to make it easier to understand) for higher (non-species taxon) taxonomic rank mutation spectra analysis we have suggested and implemented “uncertainty principle” approach consisting of: 1) consideration of the probabilities of all alternative ancestral nucleotides at each multiple alignment position on each inner node of the phylogenetic tree; and 2) additionally, consideration of the depth of the branches of a given phylogenetic tree by multiplying the probabilities of ancestral states of node i by the y_i/d ratio, where y_i is the distance

from node i to closest leaf (sum of branch lengths in the path from leaf to node i), d is the geometric mean of distribution of distances from ingroup root to all leaves. Our algorithm is based on an exhaustive analysis of the probabilities of all ancestral states obtained at each node of a given phylogenetic tree using IQ-TREE 2.2.0 (23, 30–32). It is of importance that these probabilities can be calculated from non-time-reversible Markov models that can significantly improve the final results (20, 21).

To test the effectiveness of our “uncertainty principle” approach (see above) to filtering out errors/inaccuracies in ancestral reconstruction, on Figure S2 and Figure S3 we show pairwise comparisons (by cosine similarities) of mutation spectra obtained by various approaches on identical branches of mammalian ND1 and CYTB trees (the short data description of these datasets is given in Fig. S4). We compared our fast approach (named “uncertainty principle” normalization) with 1) the best but much slower (~1000 times slower) analogous software available for general ancestral reconstruction, the PASTML software (33), which allows, on the basis of combination of marginal posterior probabilities approximation (MPPA) and the Brier score, to weight the probability of ancestral nucleotides and, consequently, to determine the most optimal discrete (non-alternative) ancestral states, and 2) the fastest standard and simplified (maximum a posteriori) discrete (non-alternative) MAP approach to reconstruct ancestral states, which reconstructs the ancestral nucleotide just like the nucleotide with the greatest local probability for the alignment site and the tree node (34). To simplify the comparison of different approaches, we used time-reversible GTR (Tavaré S (1986). *Lectures on Mathematics in the Life Sciences*. 17: 57–86) and non-reversible RY10.12 model (21) (the best fitted model to our datasets) in our reconstruction by IQ-TREE 2.2.0. For the PASTML we used the following parameters: MPPA, the Brier score aware reconstruction based on the HKY model (35). To be as coherent with the neutrality of the mutations as possible and at the same time to work with most reliable data, we have implemented a search for optimal threshold parameters on both CYTB and ND1 vertebrate data sets. First, we unambiguously showed that only classes 5 and 6 of synonymous site-rate heterogeneity (data categorized into 6 site-rate classes) have similar mutation spectra, but other (more conservative) synonymous site-rate heterogeneity classes have strongly different mutation spectra (Fig. S5). Such a difference in mutation spectra can be driven by negative natural selection. Second, we compared the mutation spectra, computed based on 10 classes of mutation probabilities, and obtained an unambiguous result that mutations with probabilities < 0.3 give the most strongly distinct mutation spectra, regardless of whether all site-rate heterogeneity classes were taken into analysis or only the most varying site-rate classes 5 and 6 were analyzed (Fig. S6). Furthermore, the inclusion of mutations with probabilities < 0.3 introduces a significant variation between our approach and PASTML (Fig. S2; Fig. S3). Therefore, we recommended to filter out all substitutions with probabilities below 0.3, which are usually located on the deep internal branches of the phylogenetic tree, due to inefficiency of deep inner ancestral states reconstruction. Third, we show that the inclusion of all synonymous site-rate heterogeneity classes into the mutation spectra computation significantly reduced the cosine similarities between our and PASTML mutation spectra for synonymous sites as compared to cosine similarities between our and PASTML mutation spectra for all (synonymous and non-synonymous) sites (Fig. S7). Fourth, we additionally demonstrated that more strict exclusion of the data deficient branches (less than 20 mutations per branch and less than 16 from 192 types of substitutions in trinucleotide context) reduced the cosine similarities between our and PASTML mutation spectra for synonymous sites too (Fig. S8). Thus, we strongly recommend users to remove ancestral states with lowest probabilities from all computations ($p < 0.3$), and also remove all synonymous sites, which have significant evolutionary restrictions on variation (for gamma distribution of site-rate heterogeneity categorized into 6 rate categories, we recommended use rate categories 5 and 6 only) in order to be as consistent as possible with the neutrality of the mutations and simultaneously to work with data with the utmost confidence.

In general, we demonstrated the effectiveness of our “uncertainty principle” normalization approach with the defined thresholds (see above) applied, we showed that this fast computational approach (~1000 times faster than PASTML) provides similar mutation spectra result as compared to one reconstructed by the best but slowest general ancestral approach to date, PASTML.

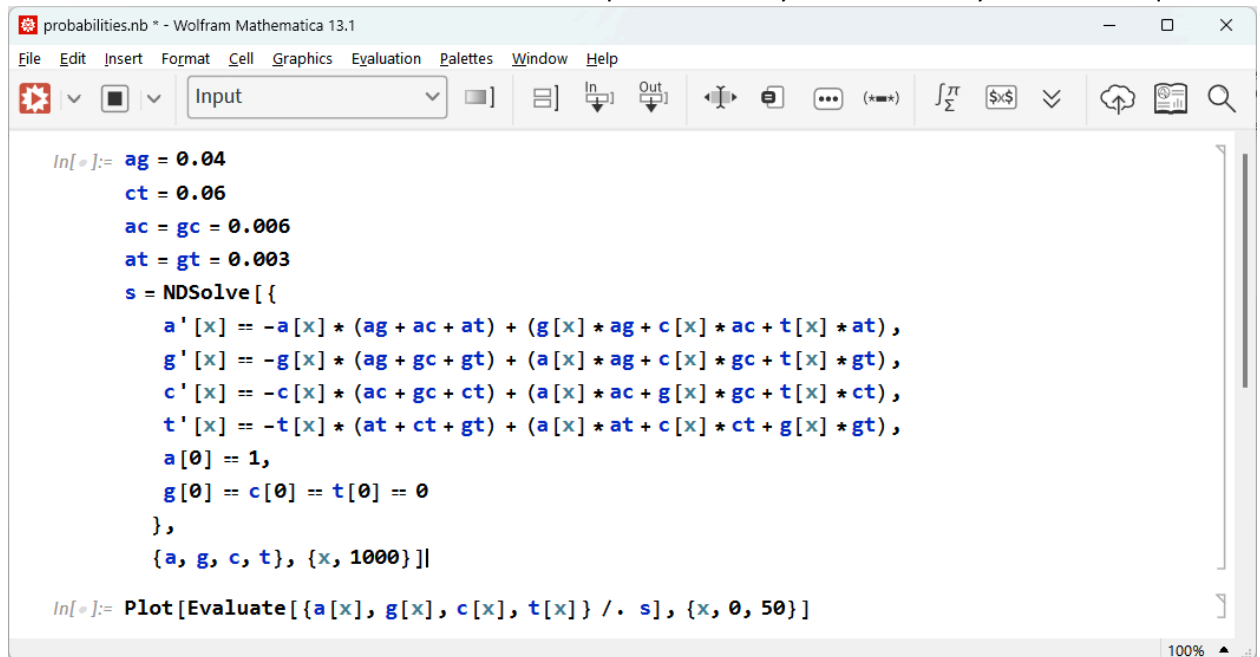
For example, the evolution of DNA can be simply mathematically formalized as a system of ordinary differential equations showing changes of all nucleotide frequencies (A, G, C and T) in time; here in a time-reversible manner:

$$\begin{cases} f'_A(t) = -f_A(t)\mu_A + \sum_{x \neq A} f_x(t)\mu_{Ax} dt, \\ f'_G(t) = -f_G(t)\mu_G + \sum_{x \neq G} f_x(t)\mu_{Gx} dt, \\ f'_C(t) = -f_C(t)\mu_C + \sum_{x \neq C} f_x(t)\mu_{Cx} dt, \\ f'_T(t) = -f_T(t)\mu_T + \sum_{x \neq T} f_x(t)\mu_{Tx} dt, \end{cases} \text{ where } \mu_{yx} = \mu_{xy} \ (y \neq x)$$

For example, in matrix form it could be rewritten as:

$$Q = \begin{bmatrix} -\mu_A & \mu_{AG} & \mu_{AC} & \mu_{AT} \\ & -\mu_G & \mu_{GC} & \mu_{GT} \\ & & -\mu_C & \mu_{CT} \\ & & & -\mu_T \end{bmatrix} = \begin{bmatrix} -0.049 & 0.04 & 0.006 & 0.003 \\ & -0.049 & 0.006 & 0.003 \\ & & -0.072 & 0.06 \\ & & & -0.072 \end{bmatrix}.$$

We can use Wolfram Mathematica software to simple solve this system of ordinary differential equations:



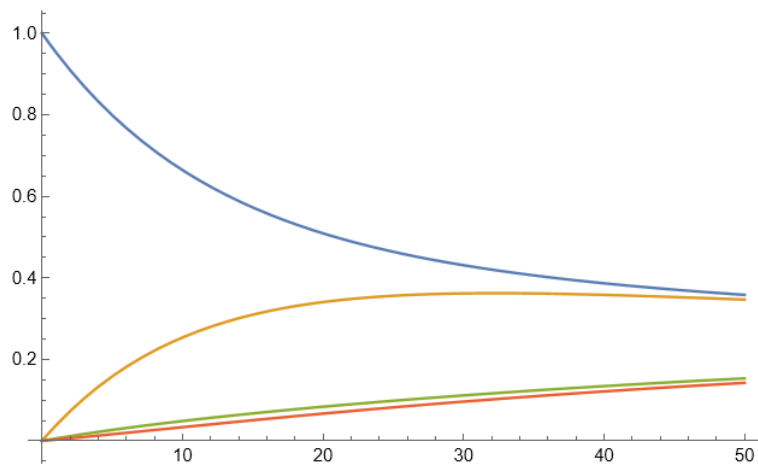
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probabilities.nb - Wolfram Mathematica 13.1
File Edit Insert Format Cell Graphics Evaluation Palettes Window Help
Input

In[ ]:= ag = 0.04
      ct = 0.06
      ac = gc = 0.006
      at = gt = 0.003
      s = NDSolve[{
        a'[x] == -a[x] * (ag + ac + at) + (g[x] * ag + c[x] * ac + t[x] * at),
        g'[x] == -g[x] * (ag + gc + gt) + (a[x] * ag + c[x] * gc + t[x] * gt),
        c'[x] == -c[x] * (ac + gc + ct) + (a[x] * ac + g[x] * gc + t[x] * ct),
        t'[x] == -t[x] * (at + ct + gt) + (a[x] * at + c[x] * ct + g[x] * gt),
        a[0] == 1,
        g[0] == c[0] == t[0] == 0
      },
      {a, g, c, t}, {x, 1000}]

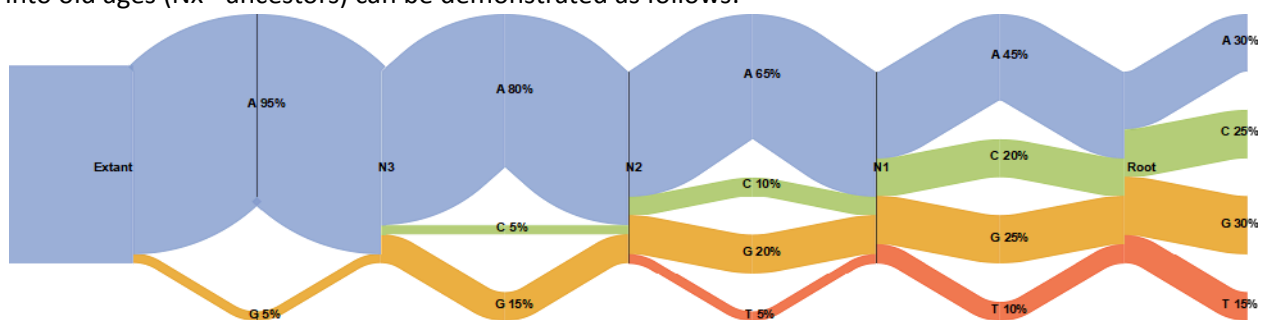
In[ ]:= Plot[Evaluate[{a[x], g[x], c[x], t[x]} /. s], {x, 0, 50}]
  
```

As a result, we can clearly demonstrate that the more the time of evolution we analyzed, the less the expected differences in nucleotide frequencies we have:



Time-related ancestral nucleotide probabilities according to the substitution model that showed above; from contemporary time (extant nucleotide is A) into old ages: A – blue; G – orange, C – green, T – red colors.

Thus, the scheme of blurring probabilities for ancestral sequence states (it may also represent an imprecision of sequence alignment near the root (36)), from contemporary time (extant nucleotide is A) into old ages (Nx - ancestors) can be demonstrated as follows:



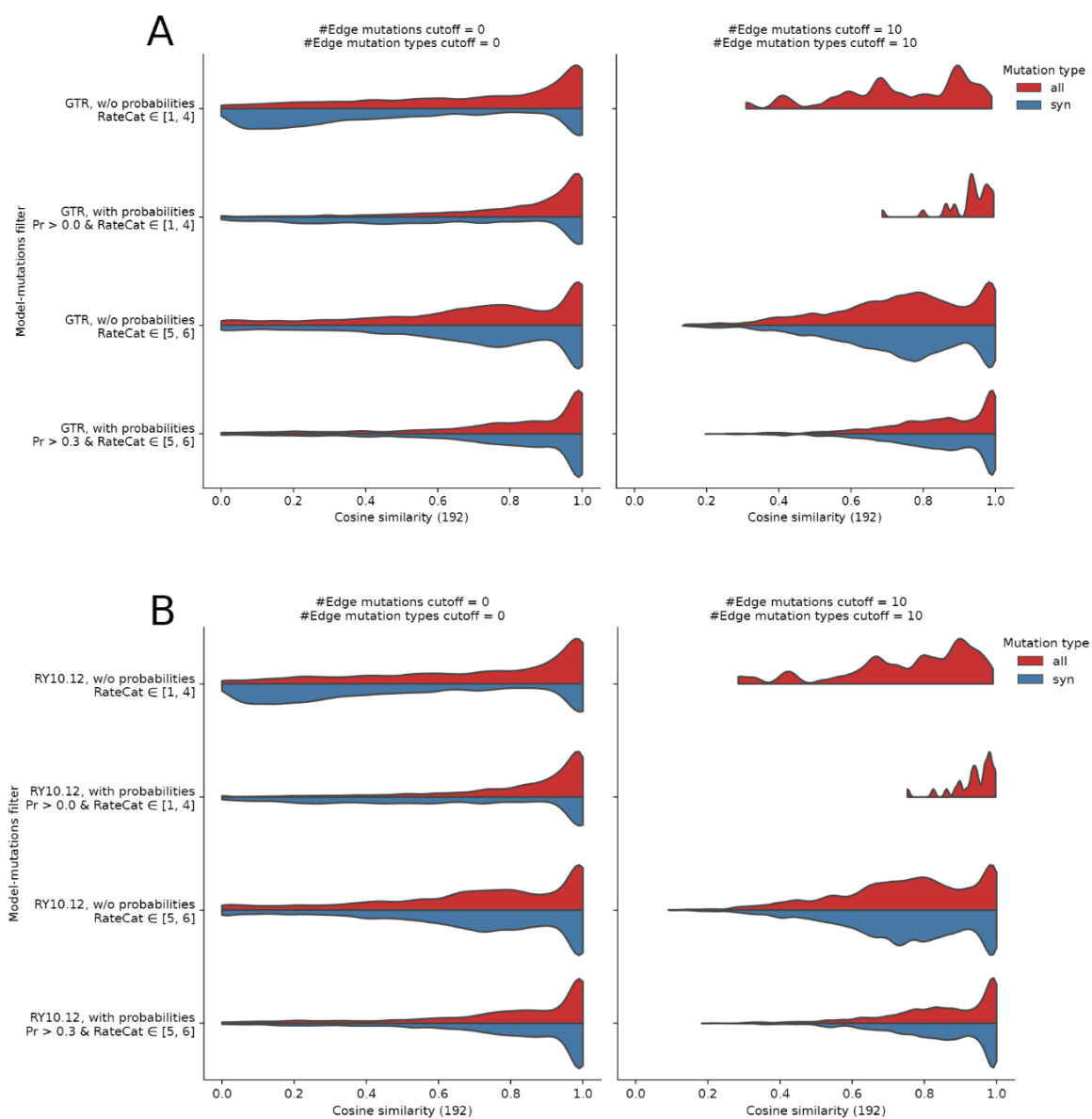


Figure S2. The distributions of pairwise cosine similarities of ND1 Vertebrate mutation spectra computed by PASTML (MPPA, the Brier score aware reconstruction based on HKY model) versus our two approaches: under GTR time reversible model (upper panel) and RY10.12 time non-reversible model (bottom panel). The distributions of pairwise cosine similarities from comparison of mutation spectra for all branches and mutations occurred in alignment sites of two rate categories, [1..4] and [5,6], demonstrated on the 1,2 and 3,4 rows, respectively. Left column shows the distributions of all 192-component mutation spectra, no filters applied. Right column shows the distributions of the 192-component mutation spectra based on tree branches having at least 10 mutations of 10 types. Odd rows demonstrate comparison of our simple ancestral states inference (MAP reconstruction of single/best ancestral state). Even rows demonstrate comparison of mutation spectra considering probabilities of all possible ancestral states.

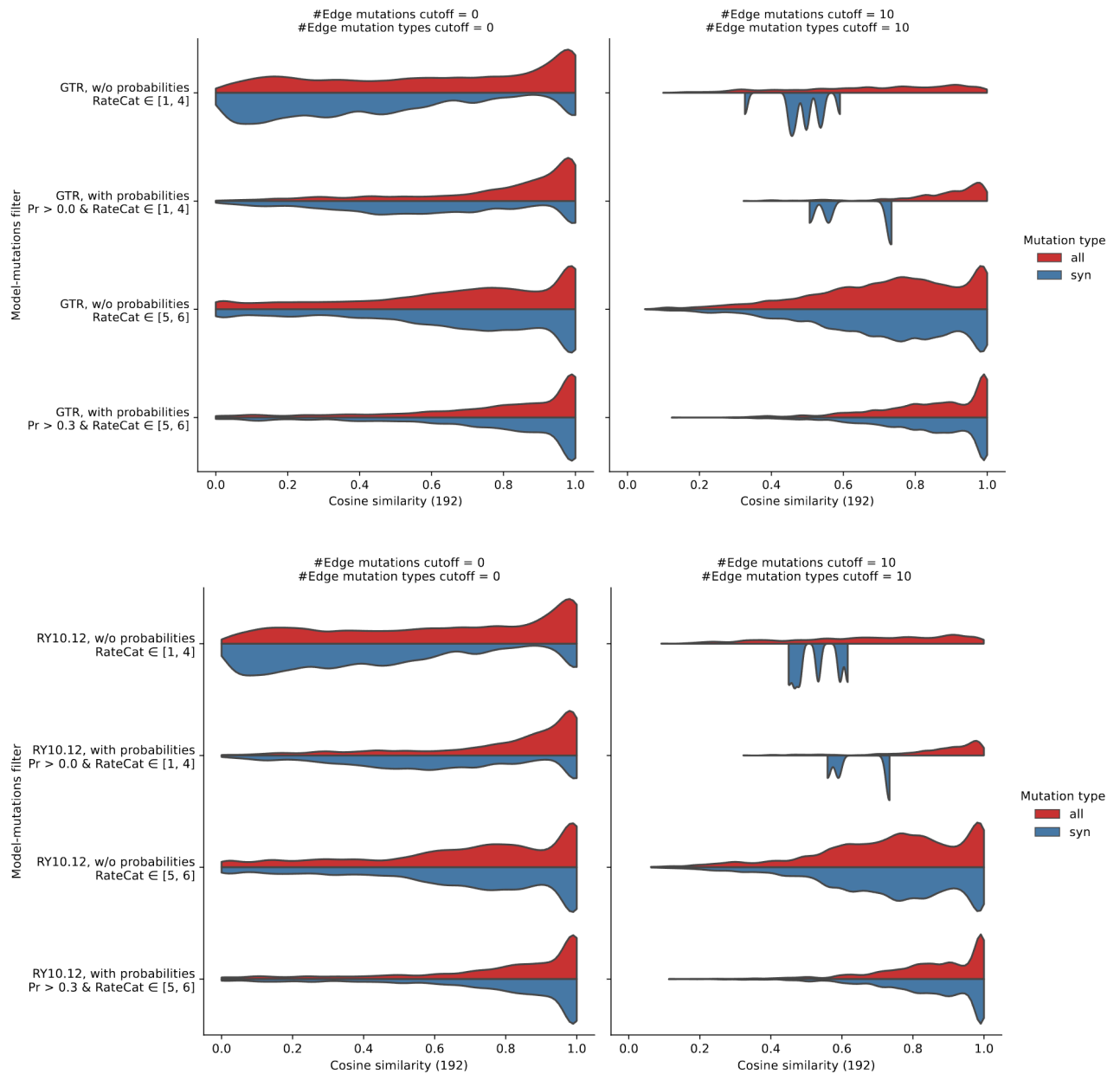
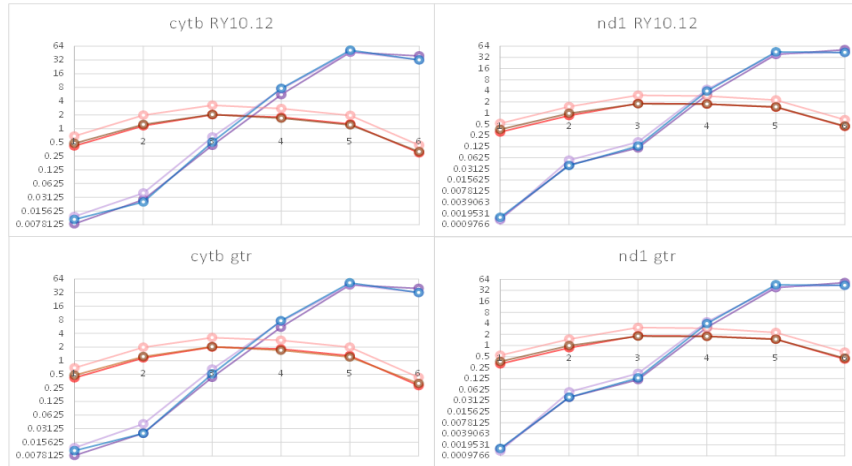


Figure S3. The distributions of pairwise cosine similarities of CYTB Vertebrate mutation spectra computed by PASTML (MPPA, the Brier score aware reconstruction based on HKY model) versus our two approaches: under GTR time reversible model (top panel) and RY10.12 time non-reversible model (bottom panel). The distributions of pairwise cosine similarities from comparison of mutation spectra for all branches and mutations occurred in alignment sites of two rate categories, [1..4] and [5,6], demonstrated on the 1,2 and 3,4 rows, respectively. Left column shows the distributions of all 192-component mutation spectra, no filters applied. Right column shows the distributions of the 192-component mutation spectra based on tree branches having at least 10 mutations of 10 types. Odd rows demonstrate comparison of our simple ancestral states inference (MAP reconstruction of single/best ancestral state). Even rows demonstrate comparison of mutation spectra considering probabilities of all possible ancestral states.

A

CYTB	0.23	0.21	0.09	0.06	0.09	0.21	0.11
ND1	0.19	0.22	0.09	0.06	0.08	0.19	0.16
	0	1	2	3	4	5	6

B



C

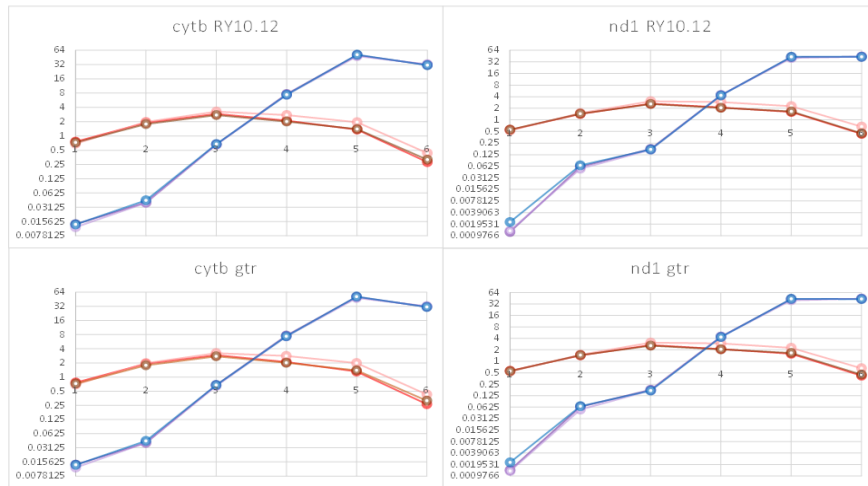


Figure S4. Variability of alignment sites in mammalian cytb and nd1 mitochondrial genes. (A) The fraction of alignment sites characterizing by 0 - 6 rate heterogeneity categories. Six rate categories (from 0, totally conservative, to 6, totally variable) of gamma distribution were used. (B) The percent (Y-axes) of substitution types and rate heterogeneity classes (X-axes) across alignment sites, all probabilities of mutations. Two substitution types were shown by different colors: synonymous substitutions by purple/blue, and non-synonymous substitutions by red/brown. Saturated and light red and purple color tones show “uncertainty principle” normalization of ancestral states (considering and weighting probabilities of all possible ancestral states) and simple ancestral states inference (MAP reconstruction of single/best ancestral state), respectively. Blue and brown colors show PASTML results (MPPA, the Brier score

aware reconstruction based on HKY model). (C) The percent (Y-axes) of substitution types and rate heterogeneity classes (X-axes) across alignment sites, probabilities of mutations > 0.3. Set of colors like in (B).

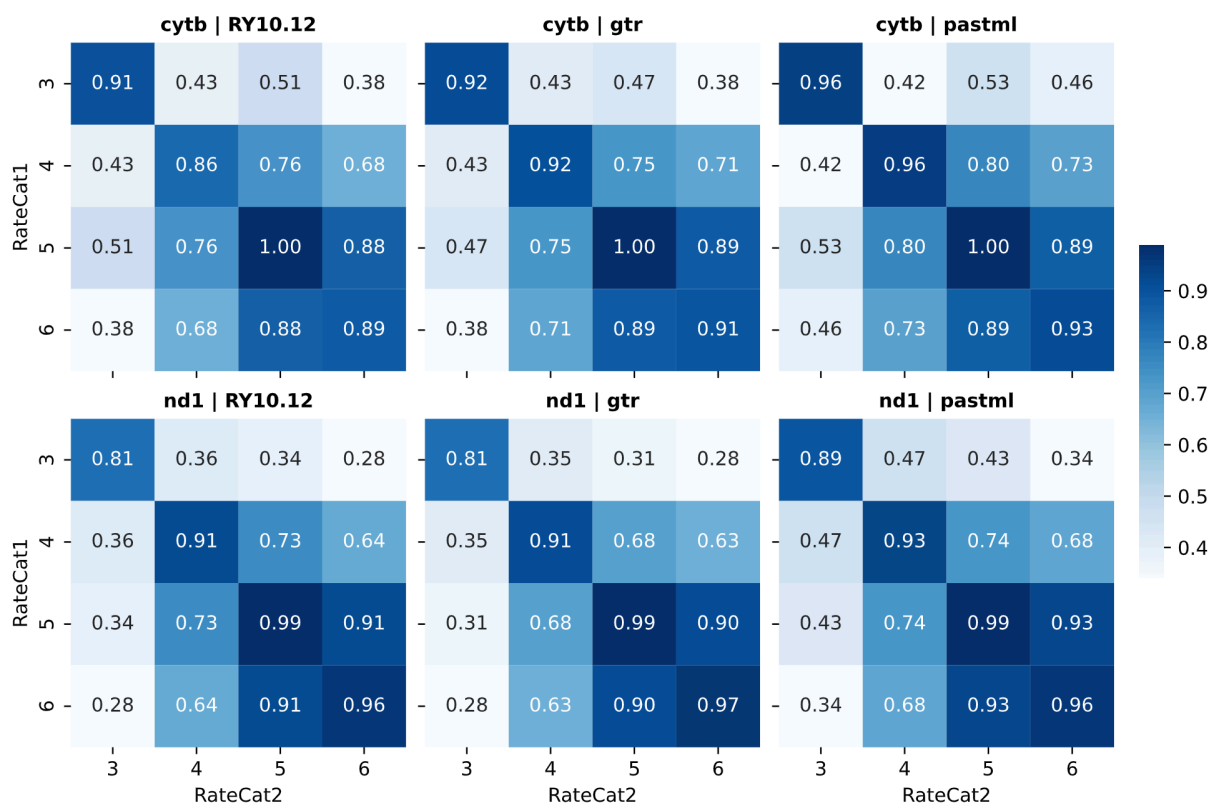


Figure S5. Pairwise comparisons of mutation spectra (by cosine similarities) computed on the basis of the last four classes of site-rate heterogeneity (invariable class 0 and conservative classes 1 and 2 were omitted due to a lack of data). The median cosine similarities obtained by 50% jackknife were shown.

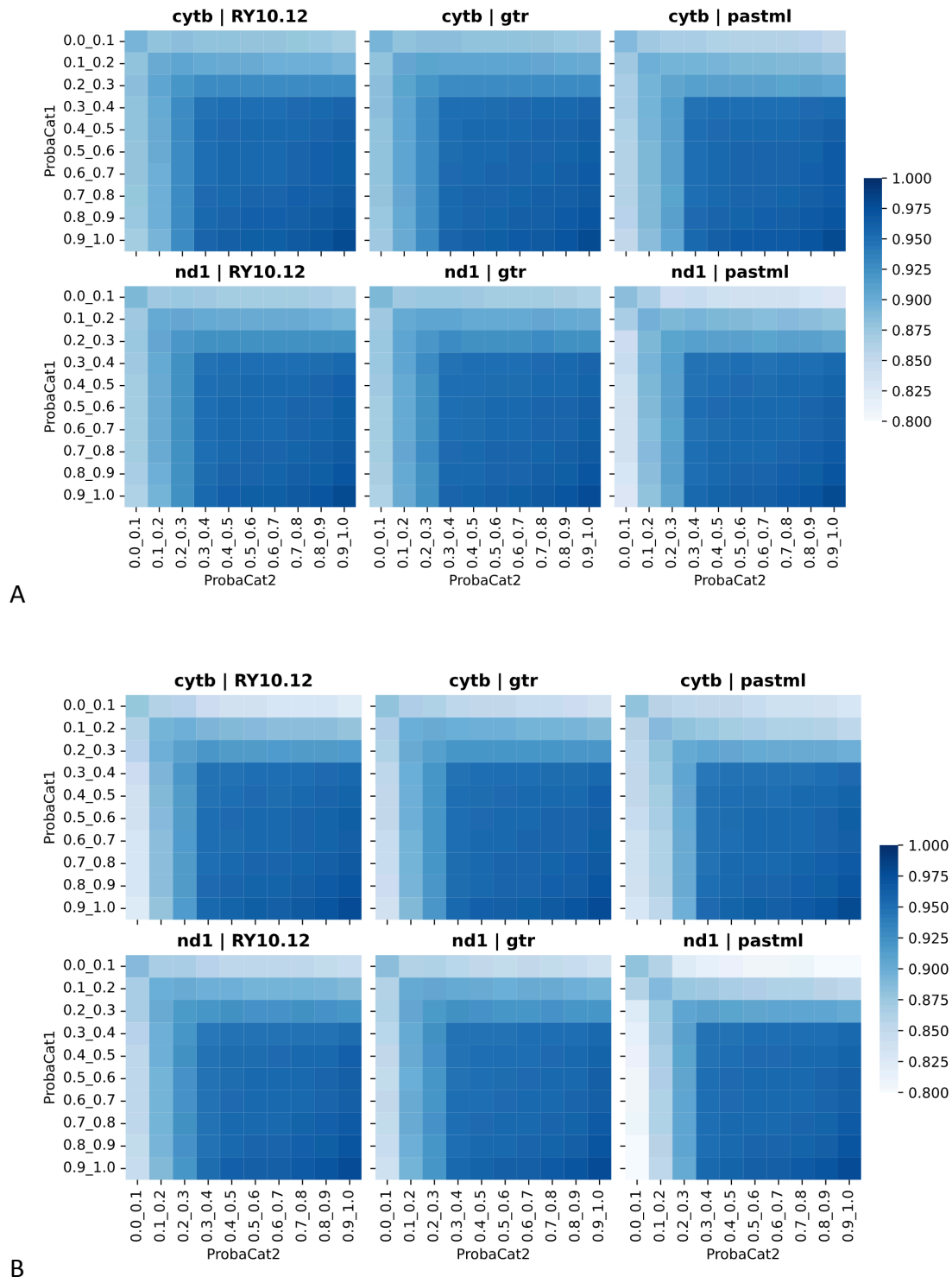


Figure S6. Pairwise comparisons of mutation spectra (by cosine similarities) computed on the basis of the 10 classes of mutation probability classes. (A) Computation based on all site-rate heterogeneity classes, (B) Computation based on the 5 and 6 site-rate heterogeneity classes. The median cosine similarities obtained by bootstrap were shown. The median cosine similarities obtained by 50% jackknife were colorized.

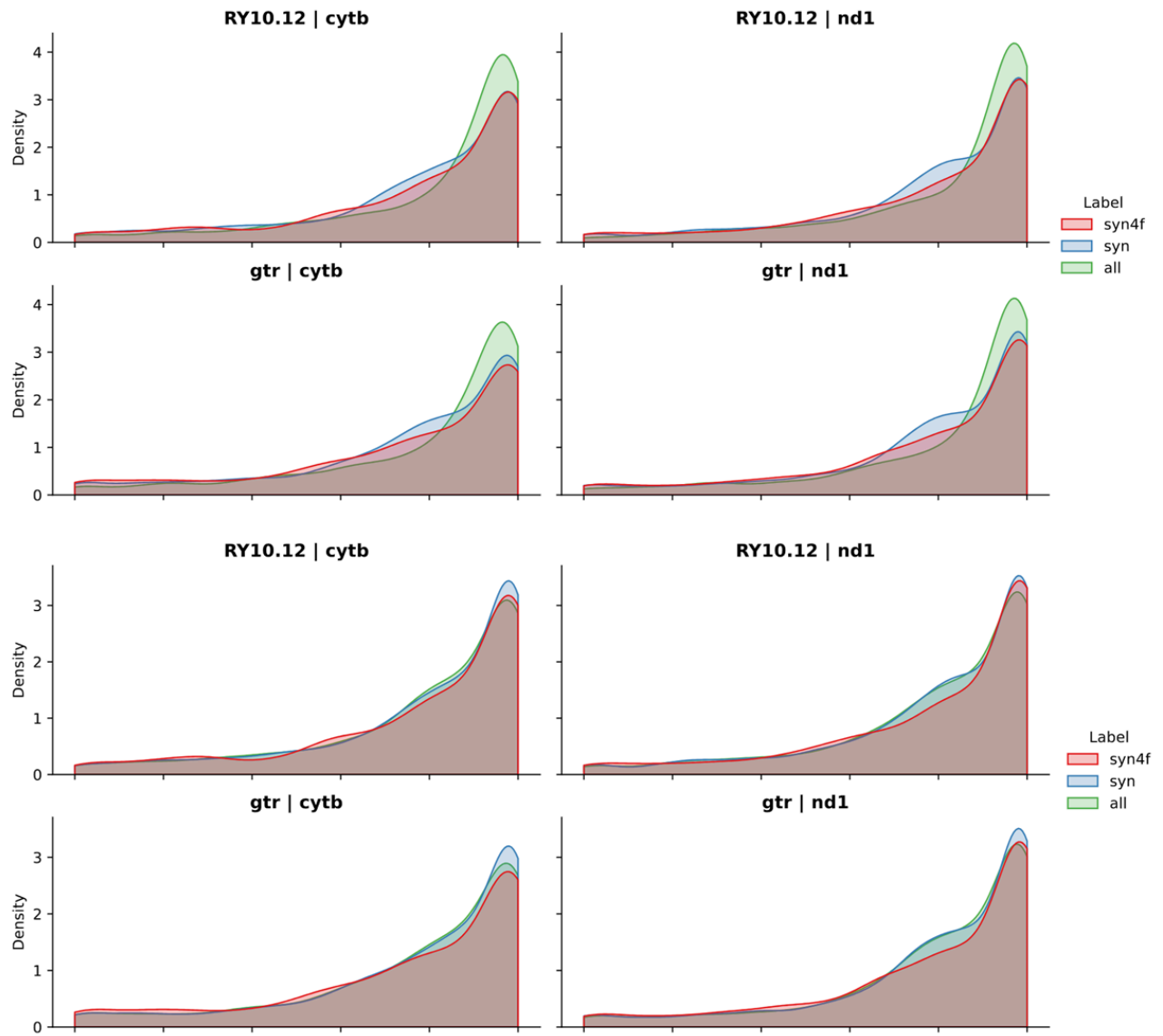


Figure S7. The effect of inclusion of all synonymous site-rate heterogeneity classes into the mutation spectra computation on cosine similarities between our and PASTML mutation spectra. Upper 4 plots, all site-rate heterogeneity categories under consideration. Lower 4 plots, only 5 and 6 site-rate heterogeneity classes analyzed.

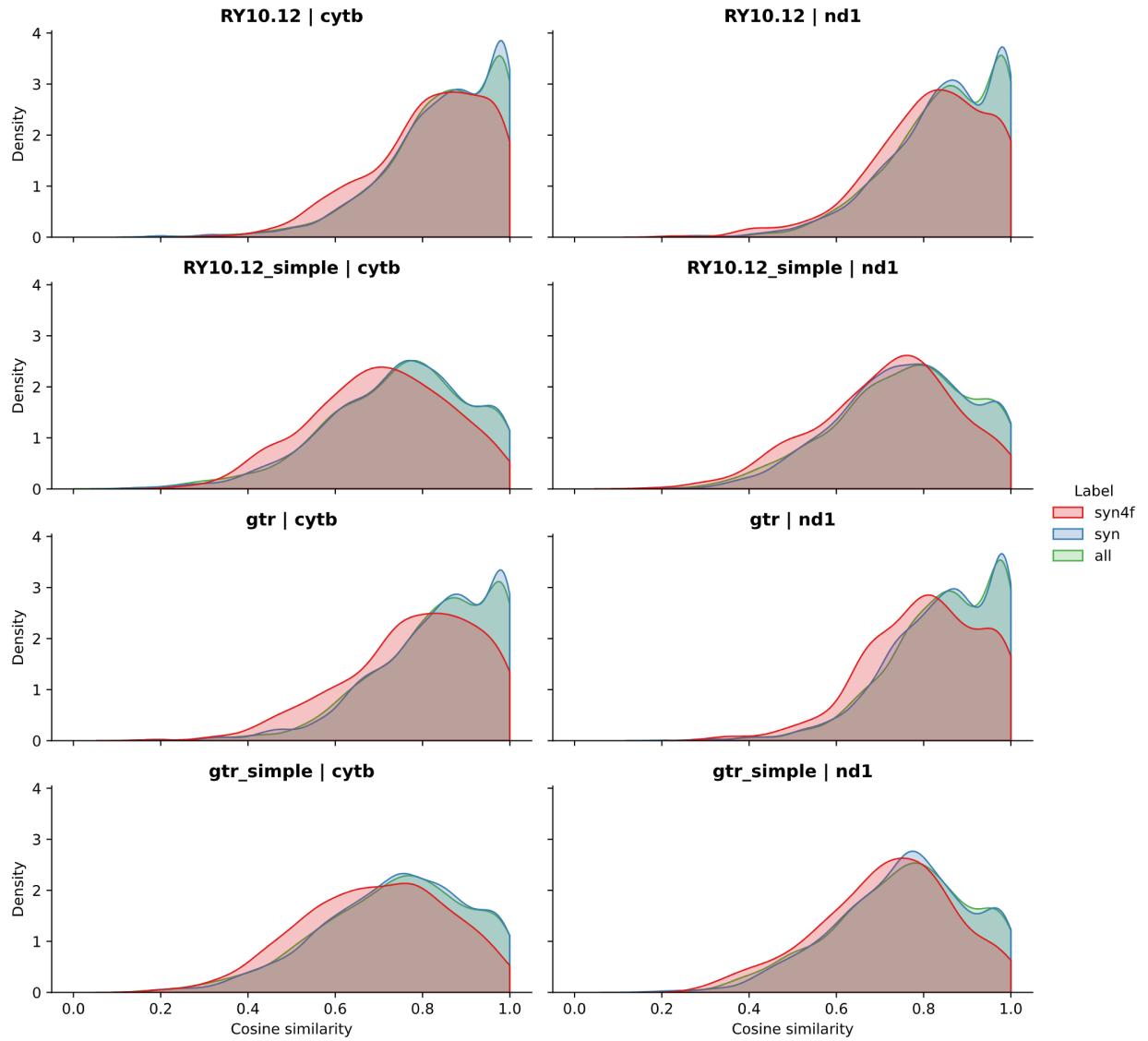


Figure S8. The distributions of pairwise cosine similarities of CYTB (left column) and ND1 (right column) Vertebrate mutation spectra (as in Fig. S2, Fig. S3), the probabilities of mutations >0.3 , only 5 and 6 site-rate heterogeneity classes, short branches with data deficiency (less than 20 mutations per branch and less than 16 from 192 types of substitutions in trinucleotide context) have been omitted.

Supplementary Note 3. The equality of mutation count normalization schemas

In cancer studies to build mutation spectra, it is common to normalize the frequencies of observed mutations using the trinucleotide frequency in the human exome or genome (37) or using the trinucleotide frequency of genes under study. On a comparative-species scale differences in nucleotide content of species could significantly affect the rate of observed mutations (for example in AT rich genome the majority of observed mutations would be from A and T even if their rates are identical to mutations from C and G) and thus should be taken into account carefully. To do this, we normalize the frequencies of the mutations observed in the trinucleotide content (taking into consideration ancestral states probabilities) using two alternative schemes: 1) using the frequency of trinucleotides (trinucleotides centered on synonymous sites) of all sequences (ancestral and existing) of the phylogeny analyzed, we do so for compatibility with cancer data stored in the COSMIC database (38), 2) using the frequency of trinucleotides (trinucleotides centered on synonymous sites excluding conservative ones) from all possible synonymous alternatives of all sequences (ancestral and existing) of the phylogeny analyzed, we make it on the basis of the simple evolutionary logic of Nei and Gojobori (39). Comparing these two normalization procedures demonstrates that the results obtained are robust to variation in data normalization (Fig. S9).

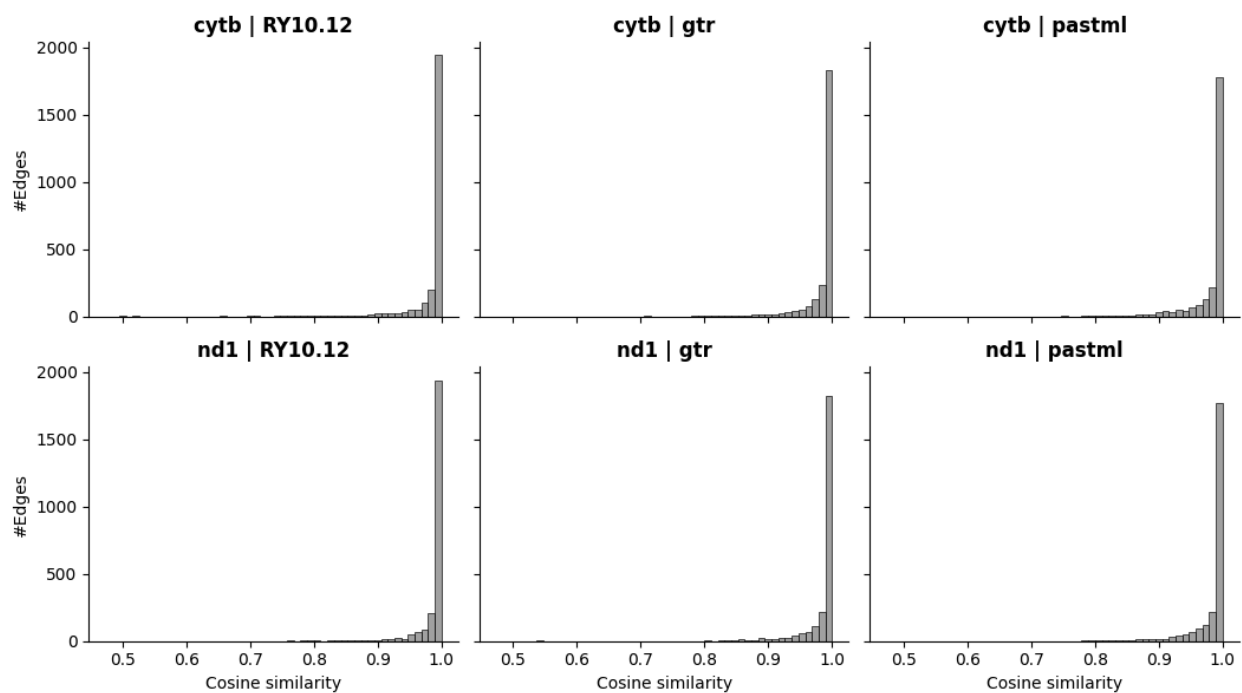


Figure S9. The distributions (for each tree branch) of pairwise cosine similarities between two mutation spectra normalisations : by the frequency of trinucleotides, in a COSMIC - like way, and by the frequency of trinucleotides from all possible synonymous alternatives, in Nei and Gojobori - like way.

Supplementary Note 4. Comparison between observed nearly neutral mutational spectrum and fully neutral mutational spectrum expectations

Currently, there is a broad range of codon evolution Markov models known as MutSel models that specifically separate the effects of neutral mutagenesis from selection based on the Fisher-Wright mutation selection framework (40–49). In this respect, we implemented an additional custom pipeline feature, we can compare the observed nearly neutral mutational spectrum with the predicted/simulated fully neutral predicted spectra. To do so we used Pyvolve package (50).

Due to the high computational demands for this step, we implemented this step only in the Singularity container of the NeMu pipeline, for advanced users only. We implemented the step based on the most powerful type of MutSel Markov model of codon evolution that made it possible to have a selective coefficient even on each synonymous change (51) implemented in Pyvolve package (50). This package supports site-wise heterogeneity (setting evolutionary variability even for each alignment site separately), phylogenetic branch lengths heterogeneity, the specification of a custom rate matrix for simulation, and custom set of states to evolve as well. We modified the Pyvolve package to add support for different genetic codes. This modified package version used in our simulations is available at <https://github.com/kpotoh/pyvolve>. We performed MutSel simulations in accordance with the procedure described in (52) (https://github.com/sjspielman/mutsel_benchmark/blob/master/simulation/simulate_alignments.py). The initial step was to assign mutation rates based on the 12-component mutation spectrum of synonymous four-fold mutations that we observed. We did this individually for every gene under analysis. Step two involved analysis of multiple codon alignments for each gene to determine the codon frequencies for each alignment site, which were then incorporated into the simulations. Third, to simulate the neutral evolution of ancestral sequences of each studied gene, we utilize observed phylogenetic trees that have unequal branch lengths. As a result of these simulations, it is possible to compare the observed and the expected/simulated mutational 192-component spectra for each analyzed gene directly (for instance, by comparing the distribution of simulated/expected mutation counts for every trinucleotide against the observed mutation count). This can show the existence of potential selective pressure on specific mutated DNA motifs/trinucleotides. In Figure S10 we compared 200 simulated neutral mutational spectra of the CYTB human gene with observed ones, it is clearly demonstrated that various substitutions in multiple DNA motifs are statistically under-represented or over-represented in the observed spectrum relative to expected neutral spectra. In summary, this could streamline the biological interpretation of observed mutation spectra.

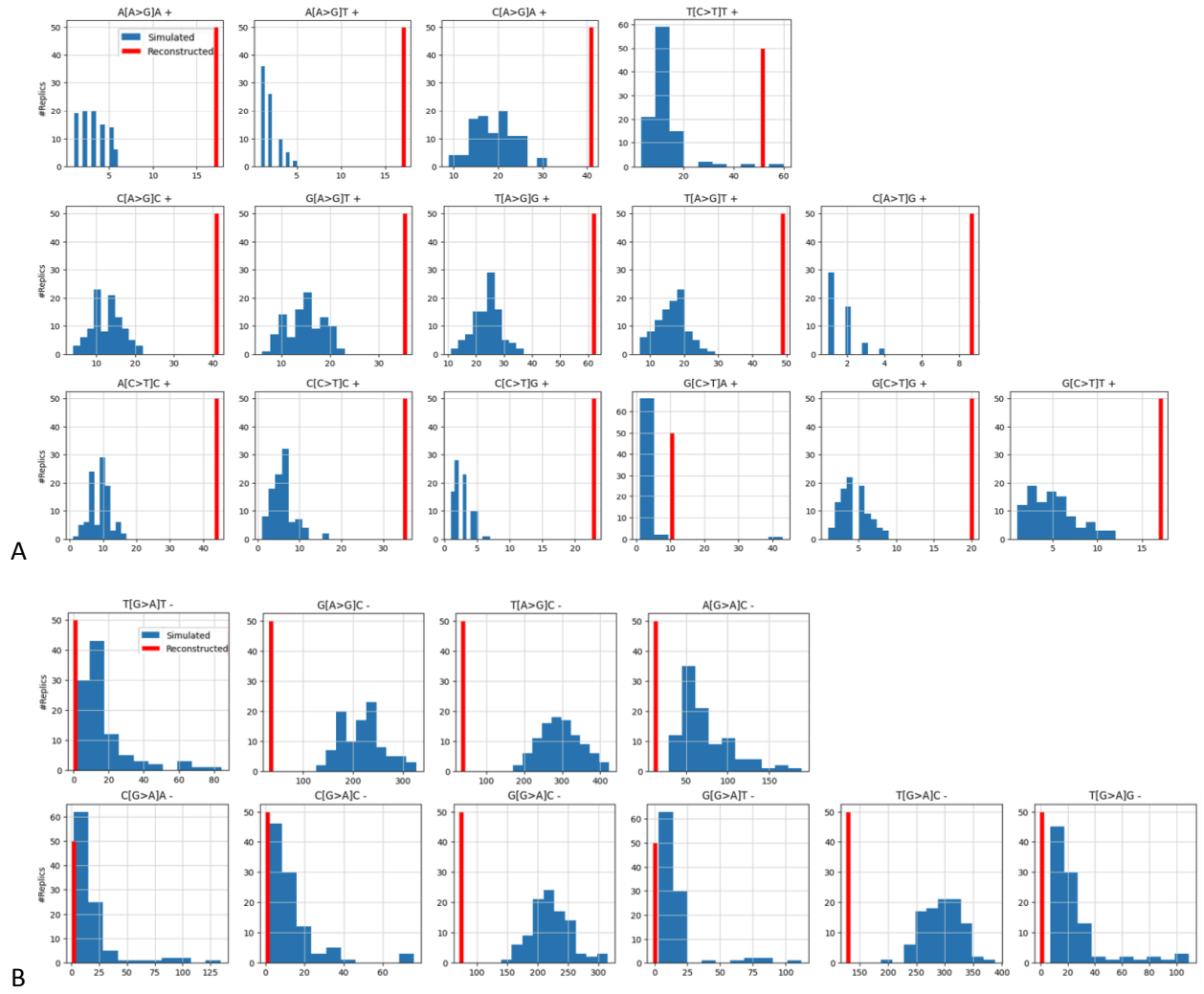


Figure S10. The comparison of 200 simulated neutral mutational spectra of the CYTB human gene with observed ones. (A) DNA motifs and substitutions statistically over-represented in the observed spectrum relative to expected neutral spectra ($p < 0.05$); (B) DNA motifs and substitutions statistically under-represented in the observed spectrum relative to expected neutral spectra ($p < 0.05$).

Supplementary Note 5. The simulation procedures for mutational spectra quality control check

Checking the validity of reconstructed ancestral states and mutation spectra in random birth-death phylogenetic simulation. In this type of simulation we took two modes: 1) based on the time-reversible GTR model of nucleotide substitution and lognormal distribution ($\mu=-2$, $\sigma=1$) of model parameters, and 2) based on non-reversible 12.12 (21) model of nucleotide substitution and normal distribution (mean=0, stddev=0.5) of model parameters. We used either the natural sequence of the human CYTB gene or the random nucleotide sequence of equal length (1140 b.p.) as the root sequence of each simulation. For all simulations we used gamma distribution of site-variation rates categorized into 6 rate categories plus invariant one (option '+G6+I' in IQ-TREE2). The empirical base frequencies (option '+F') were taken into consideration in every simulation, the base frequencies were sampled as 4 pseudo-random values from 0 to 1 and divided by sum to get actual frequencies. We also uniformly sampled the fraction of invariable sites in alignment (option '+I') from 0 to 0.5. We generated two types of random birth-death trees: shallow (100 leafs per tree) and deep (1000 leafs per tree). To normalize branch tree lengths we used the 'rlen' option with min=0, median=0.001 and max=0.01. In order to add outgroup to each tree we computed the mean length from root to leaf, multiplied it by 5 and assigned this value as outgroup branch size. For each combination of model, tree size and root sequence type we generated 100 replicates, for a total of 800 replicates.

Checking the distribution of reconstructed states and mutation spectra in true phylogenetic conditions. In this type of simulation we used actual tree topology of 4705 mammalian species [timetrees/01_step2/4705sp_mammal-time.tree file from (53)]. We also used either the natural sequence of the human CYTB gene or a random sequence (that is identical length as CYTB (1140 b.p.)) as the root sequence for each simulation. Unlike the algorithm described above, in this simulation random sequences, approximated in nucleotide composition to the natural sequences, were generated considering Shannon entropy. Only sequences whose entropy differed from the entropy of human CYTB nucleotide frequencies by no more than 0.05 were used. (Fig. S11). We also took two models, GTR and 12.12, as described above. In this simulation, in order to be as precise as possible, we 1) replaced the cosine similarities by Euclidean distances for the comparison between the simulated and observed (reconstructed) mutational spectra, 2) analyzed these distances for each of the alignment site variability class separately, and 3) analyzed synonymous mutations and all mutations separately. We used the variability class prediction of each alignment site based on gamma rate heterogeneity across alignment sites by IQ-TREE 2.2.2.7 (26, 27). Cosine similarity only takes into account the angle between vectors, while Euclidean distance takes into account their weight or magnitude. In other words, we consider the exact difference in mutation estimates on each tree branch between simulated and reconstructed by the pipeline data in these simulations. Despite the undoubted importance of Euclidean distances for mutation spectra reconstruction tests, they are not standardized as cosine similarities. To standardize Euclidean distances, or in other words, to obtain neutral expectation of Euclidean distances, we compute Euclidean pairwise distances between mutational spectra in simulated replicas. The results of reconstructions (observations) and standardized Euclidean distance distributions for synonymous mutations only demonstrated on Figures 2 and S12. It is very important that 1) the Euclidean distance distributions are multimodal for every alignment site variability class in 192 component mutational spectra, 2) only the most variable site class (cat6) have the largest and leftmost intersection between neutrally expected and observed Euclidean distance distributions, and 3) compared to a neutral expectation, the Euclidean distances of the majority of less variable classes of alignment sites are significantly biased towards zero (Supplementary Tables 1 and 2). These facts prove that only a portion of most variable synonymous substitutions can be classified as neutrally evolved cases, as expected in true natural conditions.

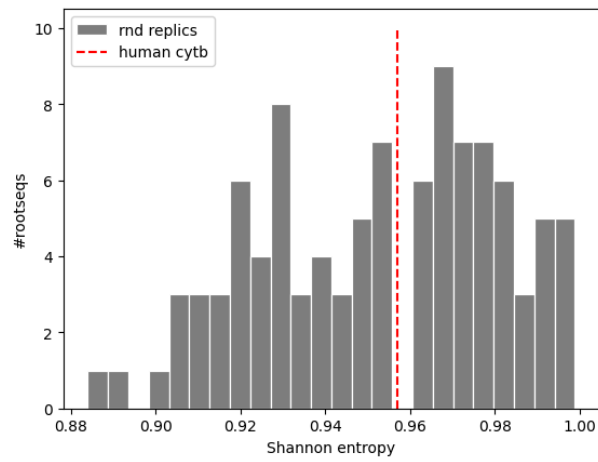


Figure S11. The distributions of Shannon entropies calculated for random root sequences in 100 replicates based on nucleotides frequencies. Red dashed line indicates entropy for the human cytb sequence. Lower entropy values denote prevalence of a single or couple of nucleotides in the sequence, i.e. not a natural scenario.

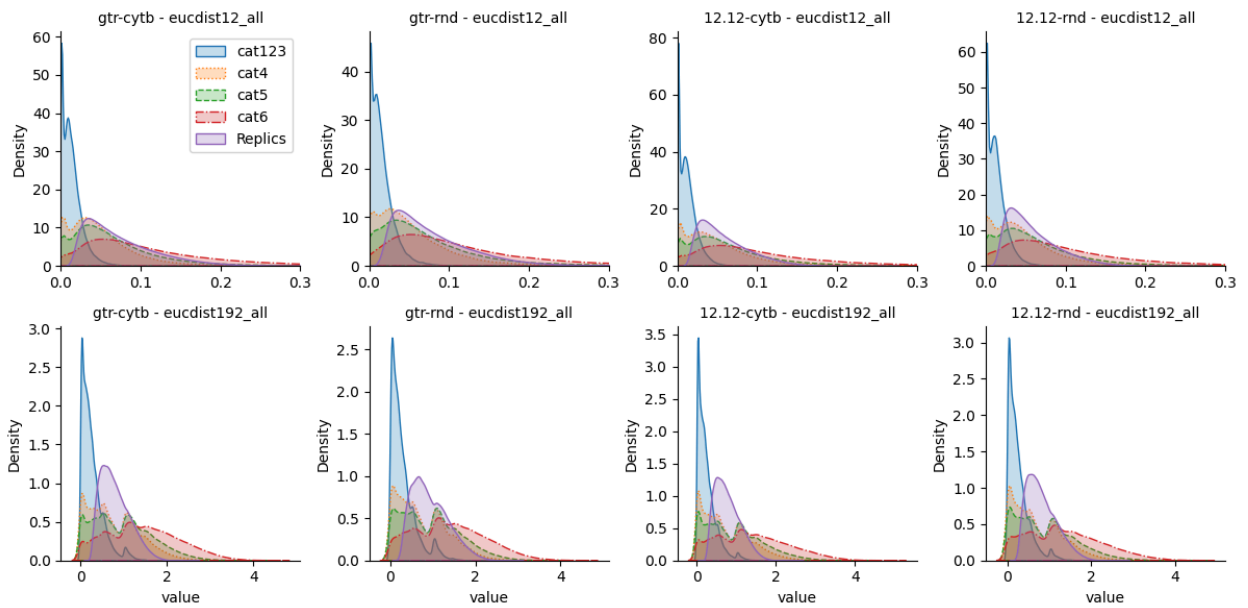


Figure S12. Testing for ancestral states and mutational spectra computational accuracy on a real tree of 4705 mammalian species: the comparison between simulated and reconstructed 12 component (top row) 192 component (bottom row) mutational spectra calculated using all mutations (synonymous and non-synonymous) based on Euclidean distances.

Supplementary Note 6. Testing the worth and meaning of reconstructing mutation spectra using a phylogenetic approach

To be as accurate as possible when choosing the strategy for mutation spectra reconstruction, we compare two alternative strategies: 1) the strategy of mutational spectrum computation that based on simple comparison of consensus sequence (either with or without correction for consistency with the common ancestral sequence; we called, majority-rule-consensus-based mutational spectrum) with each sequence in ingroup (this approach maximizes the possible number of recurrent *de novo* mutations and can significantly overestimate the mutation process, especially in cases of low and average mutation rates (54)), and 2) the strategy of mutational spectrum computation that based on probabilistic reconstruction of ancestral states in each inner tree node by IQTREE2 using GTR time reversible Markov model of nucleotide substitution (this approach based on infinite site model assumption (55) and, therefore, reduces the number of possible recurrent mutations and can underestimate the mutation process, particularly in instances of high mutation rates and mutation hotspots (56, 57); we called, standard mutational spectrum). We compared these strategies by analyzing 7017 intra-species mitochondrial datasets that were composed of the natural sequence of all mitochondrial protein-coding genes. Majority-rule-consensus-based mutational spectrum computation strategy involves reconstructing a simple majority rule consensus sequence for each intra-species mitochondrial dataset. In order to correct the majority rule consensus sequence for every intra-species mitochondrial dataset, we utilized common ancestral sequences that were reconstructed for each dataset using a phylogenetic approach (by IQTREE2). Excluding sites from the majority rule consensus that are inconsistent with common ancestral sequence significantly increases the cosine similarities between 192-component majority-rule-consensus-based and standard mutational spectra ($p < 10^{-6}$; U-test; Fig. S13). The cosine similarities between corrected majority-rule-consensus-based mutational spectrum (corrected by the consistency with common ancestral sequence) and standard mutational spectrum obtained for each dataset is additionally contrasted against the difference between tree length reconstructed by IQTREE2 and a tree length of corresponding star-like topology (Fig. S14). We obtain star-like topology for each dataset using a simple inner branch collapsing procedure with preserving the distance from the root node for each tree leaf. Thus, star-like trees were longer than the respective native phylogenetic trees. Taking this into consideration, we scaled the distance of the length of native tree from the length of the corresponding star-like tree as the $1 - [\text{length of native tree}] / [\text{length of corresponding star-like tree}]$. Fig. S14 shows that the cosine similarity values of majority-rule-consensus-based and standard mutational spectra negatively (nonlinearly) correlate with the difference between native phylogenetic trees and respective star-like topology trees. Moreover we have high statistical significance of these negative correlations ($p < 10^{-20}$) for both data subgroups (Fig. S14): (1) with sufficient number of mutations to reconstruct 192-component majority-rule-consensus-based and standard mutational spectra (cloud of dots upper $Y=0.2$; 6468 datasets or 92%; median number of mutations per dataset = 208, 25% quartile of mutations per dataset = 99, 75% quartile of mutations per dataset = 493) and (2) with lack of mutations to reconstruct such spectra (cloud of dots lower $Y=0.2$; 549 datasets or 8%; median number of mutations per dataset = 119, 25% quartile of mutations per dataset = 67, 75% quartile of mutations per dataset = 222). Thus, we clearly show that the lower the divergence of the native tree to a star-like topology, the greater the similarity between majority-rule-consensus-based and standard mutational spectra (especially in the data subgroup with sufficient number of mutations). Additionally, as anticipated (the vast majority of mtDNA have no possibility of recombination due to strict maternal inheritance), we clearly show ($p < 10^{-2}$ for every dataset with sufficient number of mutations; testing the difference of lengths of native tree from star-like tree by bootstrapping tree leaves) that the strategy of standard mutational spectrum reconstruction is much more effective at describing evolution for at least 92% of mitochondrial datasets than the reconstruction of majority-rule-consensus-based mutational spectrum.

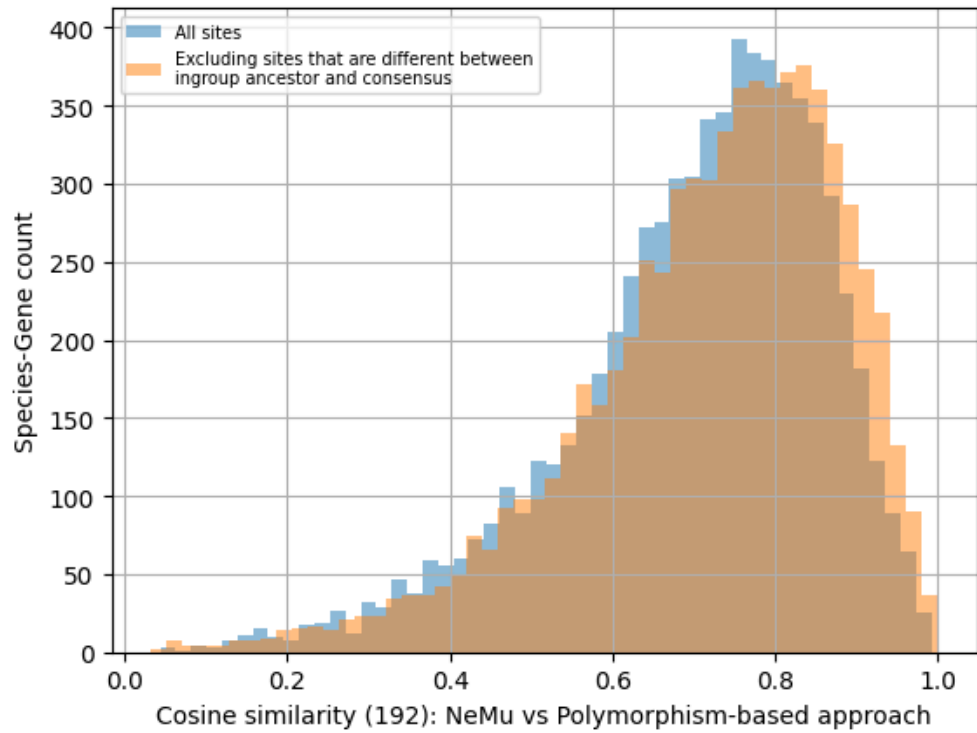
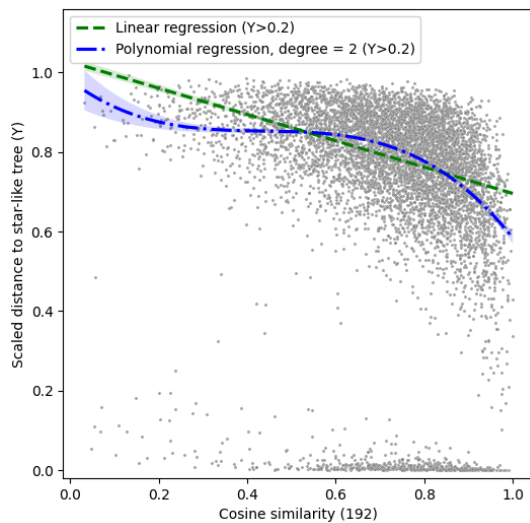


Figure S13. The correction of the major rule consensus sequence is the factor that determines the higher value of cosine similarities between 192-component majority-rule-consensus-based and standard mutational spectra.

A



B

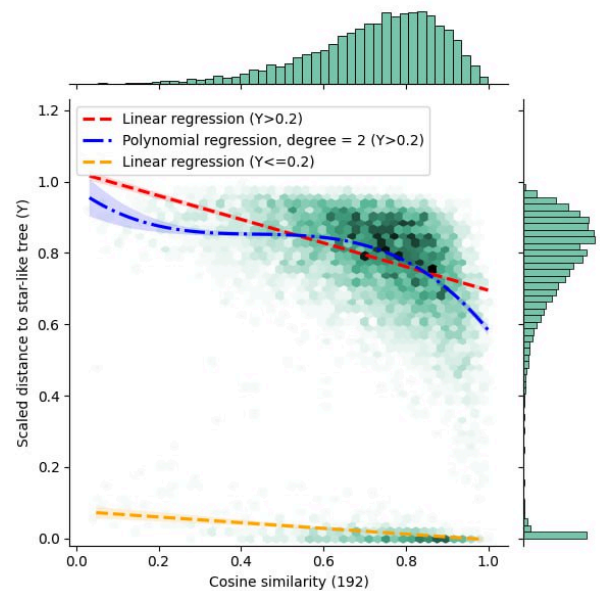


Figure S14. The comparison of cosine similarities between 192-component majority-rule-consensus-based and standard mutational spectra with the difference between tree length reconstructed by IQTREE2 and a tree length of star-like topology for each of 7017 mitochondrial dataset. Two representations of the same dataset were shown: A, dot-plot; B, density-plot.

Supplementary Note 7. The robustness of mutation spectra to changes of evolutionary model

The comparison of mutational spectra obtained on both datasets (Supplementary Files 1 and 2), ND1 and CYTB, (Fig. S15 and Fig. S16, respectively) shows that best fitted non-reversible Markov model and GTR time reversible Markov model for ancestral synonymous mutations reconstruction give us qualitatively similar estimates of mutation frequencies in trinucleotide context. First, this indicates the robustness of our interspecies mutational spectra estimates to changes from non-reversible Markov models to time-reversible. Second, it shows that if only fast approximative estimation of the mutational spectrum is desirable to the user, it is possible to use a time reversible Markov model of nucleotide substitution (e.g. GTR) rather than a non-reversible model.

Supplementary File 1. Mammalian ND1 dataset composed of 1547 sequences.

https://github.com/mitoclub/nemu-pipeline/blob/master/data/supplementary_files/nd1_mammals.fna

Supplementary File 2. Mammalian CYTB dataset composed of 1545 sequences.

https://github.com/mitoclub/nemu-pipeline/blob/master/data/supplementary_files/cytb_mammals.fna

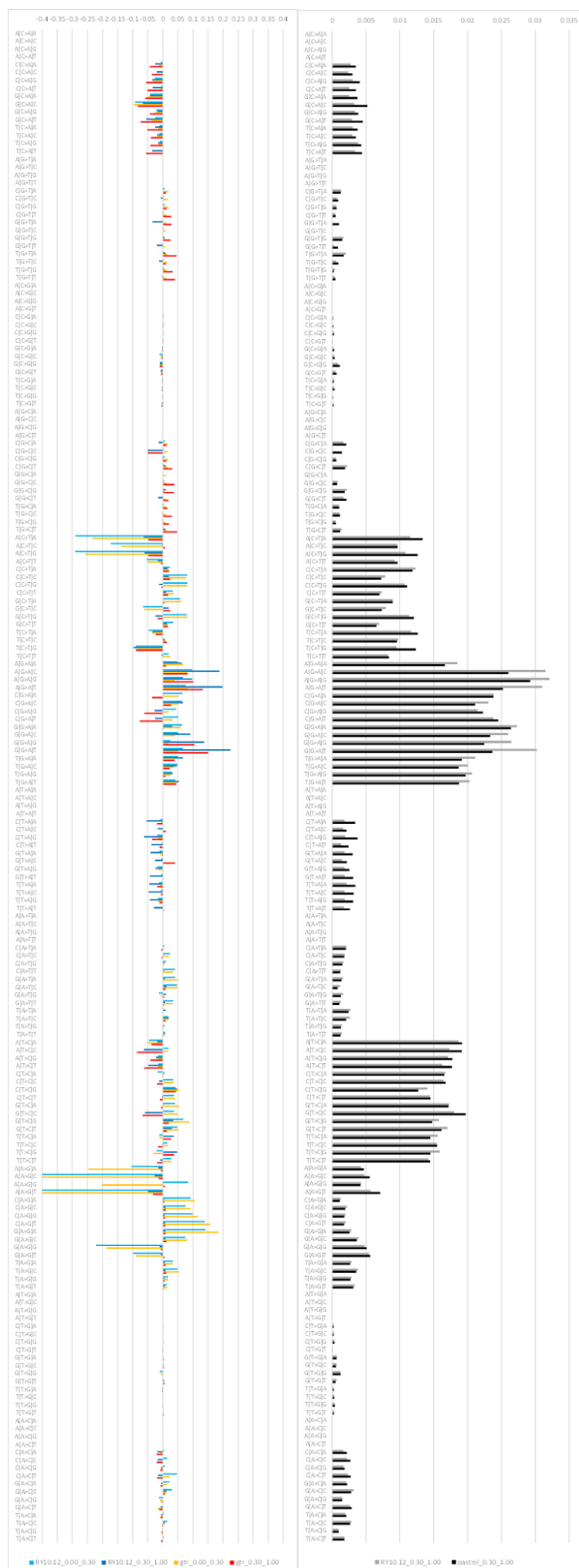


Figure S15. The Mammals CYTB mutation spectrum, RateCat [5, 6] used. Left panel, the comparison of spectra obtained from PASTML-reconstructed ancestors with our approach: red and orange bars correspond to GTR substitution model (red, $p>0.3$; orange, $p<0.3$), blue and cyan bars correspond to RY10.12 substitution model (blue, $p>0.3$; cyan, $p<0.3$). Right panel, the Mammals ND1 mutation spectra obtained by PASTML (black) and by our approach, RY10.12 substitution model (gray); both our and PASTML reconstructions based on $p>0.3$ ancestral probability class.

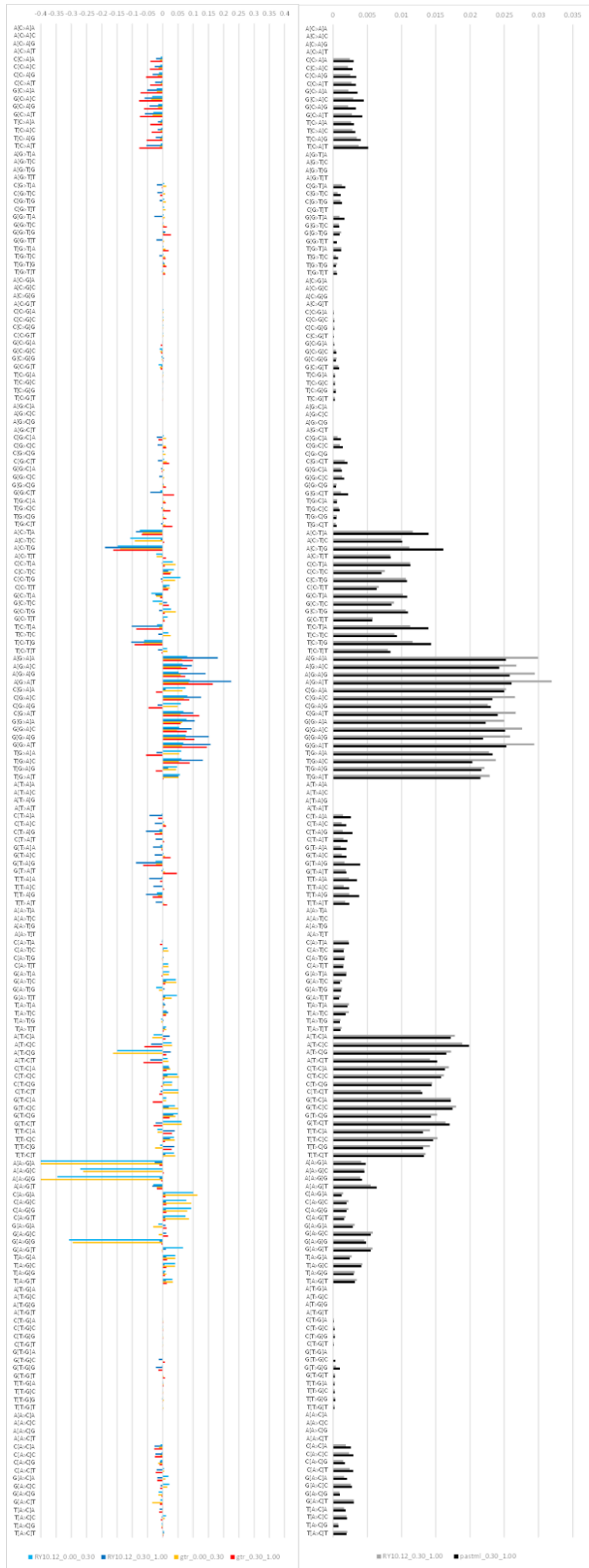


Figure S16. The Mammals ND1 mutation spectrum, RateCat [5, 6] used. Left panel, the comparison of spectra obtained from PASTML-reconstructed ancestors with our approach: red and orange bars correspond to GTR substitution model (red, $p>0.3$; orange, $p<0.3$), blue and cyan bars correspond to RY10.12 substitution model (blue, $p>0.3$; cyan, $p<0.3$). Right panel, the Mammals ND1 mutation spectra obtained by PASTML (black) and by our approach, RY10.12 substitution model (gray); both our and PASTML reconstructions based on $p>0.3$ ancestral probability class.

SUPPLEMENTARY TABLES

Supplementary Table 1. Additional species used in analysis of alignment sites variability.

<i>Hominidae</i> dataset			<i>Mus</i> dataset		
CYTB	COX1	ND1	CYTB	COX1	ND1
Gorilla_beringei	Gorilla_beringei	Gorilla_beringei	Mus_booduga	Mus_lepidoides	Mus_baoulei
Gorilla_gorilla_diehli	Gorilla_gorilla_diehli	Gorilla_gorilla_gorilla	Mus_caroli	Mus_mahomet	Mus_caroli
Gorilla_gorilla_gorilla	Gorilla_gorilla_gorilla	Homo_heidelbergensis	Mus_cervicolor	Mus_mattheyi	Mus_cookii
Homo_heidelbergensis	Homo_heidelbergensis	Pan_paniscus	Mus_cookii	Mus_mayori	Mus_famulus
Pan_paniscus	Pan_paniscus	Pan_troglodytes	Mus_crociduroides	Mus_neavei	Mus_fragilicauda
Pan_troglodytes_elliotti	Pan_troglodytes_elliotti	Pan_troglodytes_elliotti	Mus_cypriacus	Mus_nitidulus	Mus_minutoides
Pan_troglodytes_schweinfurthii	Pan_troglodytes_schweinfurthii	Pan_troglodytes_schweinfurthii	Mus_famulus	Mus_pahari	Mus_saxicola
Pan_troglodytes_troglodytes	Pan_troglodytes_troglodytes	Pongo_abelii	Mus_fernandoni	Mus_platythrix	Mus_terricolor
Pan_troglodytes_verus	Pan_troglodytes_verus	Pongo_pygmaeus	Mus_fragilicauda	Mus_poschiavinus	Mus_spretus
Pongo_abelii	Pongo_abelii		Mus_harennensis	Mus_saxicola	Mus_terricolor
Pongo_pygmaeus	Pongo_pygmaeus		Mus_haussa	Mus_spicilegus	
			Mus_triton	Mus_spretus	
			Mus_terricolor		

Supplementary Table 2. Share of edges that have euclidean distances less than 5th percentile of reference distribution.

Model	Gene	Rate Category	euclidist12_all	euclidist12_syn	euclidist192_all	euclidist192_syn
12.12	Cytb	cat123	0.7762	0.7923	0.7445	0.7857
12.12	Cytb	cat4	0.2786	0.3392	0.3204	0.5121
12.12	Cytb	cat5	0.1943	0.2931	0.2419	0.4259
12.12	Cytb	cat6	0.0868	0.1501	0.1205	0.2452
12.12	Random	cat123	0.7848	0.8185	0.7625	0.8183
12.12	Random	cat4	0.2941	0.3985	0.3482	0.5854
12.12	Random	cat5	0.2068	0.3199	0.262	0.4672
12.12	Random	cat6	0.0983	0.148	0.1358	0.25
GTR	Cytb	cat123	0.7801	0.829	0.7193	0.7738
GTR	Cytb	cat4	0.264	0.3633	0.2822	0.5042
GTR	Cytb	cat5	0.1865	0.2993	0.2078	0.399
GTR	Cytb	cat6	0.0805	0.1608	0.0984	0.228
GTR	Random	cat123	0.7738	0.8297	0.7202	0.7959
GTR	Random	cat4	0.2945	0.3902	0.3276	0.5395
GTR	Random	cat5	0.2108	0.3264	0.2433	0.4155
GTR	Random	cat6	0.0993	0.1642	0.123	0.2267

Supplementary Table 3. Share of edges that have euclidean distances greater than 95th percentile of reference distribution.

Model	Gene	Rate Category	euclidist12_all	euclidist12_syn	euclidist192_all	euclidist192_syn
12.12	Cytb	cat123	0.0001	0.0043	0.0015	0
12.12	Cytb	cat4	0.0515	0.1809	0.1318	0.0004
12.12	Cytb	cat5	0.114	0.2384	0.2327	0.005
12.12	Cytb	cat6	0.2984	0.425	0.4616	0.0449
12.12	Random	cat123	0.0001	0.0009	0.0015	0
12.12	Random	cat4	0.0483	0.1081	0.109	0.0008
12.12	Random	cat5	0.0965	0.1275	0.1998	0.0037
12.12	Random	cat6	0.2661	0.3592	0.4179	0.0662
GTR	Cytb	cat123	0	0.004	0.0014	0
GTR	Cytb	cat4	0.0142	0.084	0.1205	0.0005
GTR	Cytb	cat5	0.0461	0.137	0.2212	0.004
GTR	Cytb	cat6	0.1862	0.3017	0.4587	0.0291
GTR	Random	cat123	0.0001	0.0017	0.0005	0
GTR	Random	cat4	0.0249	0.0863	0.056	0
GTR	Random	cat5	0.056	0.1026	0.126	0.0041
GTR	Random	cat6	0.1881	0.2852	0.3312	0.0415

Supplementary Table 4. Number of successful runs of NeMu pipeline on mtDNA genes of 49461 Chordates species from MIDORI2 database. The values in the table indicate the number of species in a class for which the mutational spectrum of a given gene has been reconstructed. Last row shows the total presence of species for a given gene in MIDORI2. Last column shows the total number of mtDNA genes with reconstructed spectrum for a given class. Class names based on NCBI Taxonomy.

Class of Chordates	A6	A8	CO1	CO2	CO3	Cytb	ND1	ND2	ND3	ND4	ND4L	ND5	ND6	Total processed mtDNA genes
<i>Mammalia</i>	121	70	309	145	142	1123	153	210	97	174	81	155	118	2898
<i>Actinopteri</i>	167	56	334	31	34	1026	76	235	25	68	21	57	40	2170
<i>Lepidosauria</i>	51	21	110	7	8	435	61	425	5	337	5	12	8	1485
<i>Aves</i>	122	23	79	40	44	314	44	718	79	60	14	57	61	1655
<i>Amphibia</i>	14	5	135	8	13	242	63	181	22	52	5	20	12	772
<i>class Testudines</i>	0	0	3	0	0	40	1	1	1	16	0	2	2	66
<i>class Crocodylia</i>	1	1	3	1	1	10	1	1	1	1	1	1	1	24
<i>Hyperoartia</i>	2	1	1	0	0	9	0	1	1	0	0	0	0	15
<i>Chondrichthyes</i>	5	3	15	1	1	8	1	46	0	5	0	4	1	90
<i>Leptocardii</i>	3	0	4	2	2	3	3	3	2	2	2	3	2	31
<i>Ascidacea</i>	0	0	15	0	0	2	0	0	0	1	0	0	0	18
<i>Cladistia</i>	0	0	0	0	0	2	0	0	0	0	0	0	0	2
<i>Ceratodontiformes</i>	1	0	1	0	0	1	0	0	0	0	0	1	0	4
<i>Myxini</i>	0	0	1	0	0	0	0	0	0	0	0	0	0	1
<i>Thaliacea</i>	0	0	1	0	0	0	0	0	0	0	0	0	0	1
Total successful runs	487	180	1011	235	245	3215	403	1821	233	716	129	312	245	9232
Total species in MIDORI2	12543	12033	33506	10931	10428	33962	12684	23602	12754	14985	10350	10789	10300	208867

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